

Looping 3D Videos from a Moving Camera via Recycled 2D Tracking

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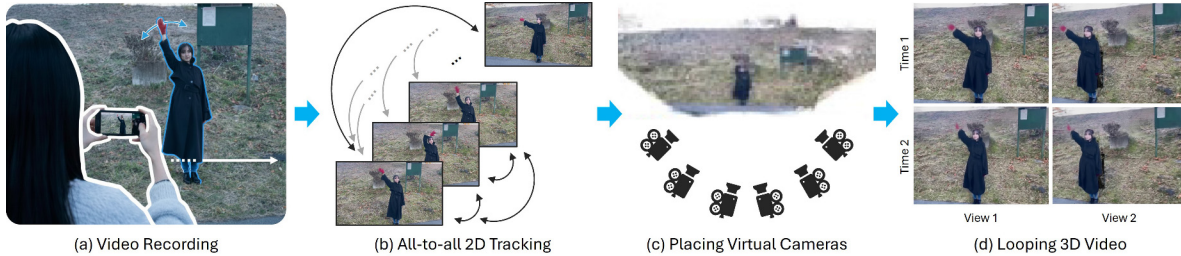


Figure 1: Generating a looping 3D video from a single moving camera. (a) A monocular video is casually captured by moving a camera. (b) Our method finds all-to-all 2D tracking by reusing pre-processed tracking data for speed. (c) It reconstructs a dynamic 3D scene to simulate multiple video cameras (d) to generate a loopy VR video.

ABSTRACT

We present a system for generating looping 3D videos from casual monocular footage captured by a single moving camera. Unlike existing looping 3D video methods that rely on a stationary multi-camera system, our approach synthesizes multi-camera views to mimic a physical multi-camera setup. Our key technical contribution is the efficient construction of all-pairs long-term 2D tracking by reusing multi-step optical flow, which avoids the need for expensive pairwise tracking. Our approach is 122.8 times faster than the pairwise approach while maintaining 98.2, 99.5, and 99.7% of the visual quality in PSNR, SSIM, and LPIPS, respectively.

Index Terms: Looping 3D videos, monocular camera, long-term 2D tracking, recycled 2D tracking.

1 INTRODUCTION

Looping 3D videos provide immersive experiences where users can freely observe dynamic scenes over extended durations without temporal discontinuities. However, generating temporally seamless loops in dynamic VR scenes is challenging due to the need to maintain consistent geometry and motion across both time and viewpoint. As a result, conventional loopable 3D video generation methods [3] rely on multiple fixed cameras to separate static and dynamic scene components.

To overcome these limitations, a promising direction is to reconstruct dynamic 3D scenes from monocular video and to render them from virtual fixed cameras within a reconstructed 4D (3D + time) space. Recent advances in dynamic scene reconstruction, such as Shape of Motion [4], make this direction feasible by jointly modeling geometry and motion from a single moving camera. However, such methods require camera poses, depth maps, and long-term 2D tracking as input, which are challenging to obtain reliably from monocular video.

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In this work, we present a pipeline that integrates MegaSaM [2] and Shape of Motion to enable loopy VR video generation directly from casually captured monocular footage. By leveraging MegaSaM to estimate camera motion and depth, and by extending its adjacent-frame optical flow into a simple and fast method for all-pair optical flow, our approach converts these features into long-term 2D tracking. This provides an efficient alternative to standard tracking-based preprocessing while supporting the generation of loopable VR videos.

2 METHOD

Overview Given a video captured with a moving camera, we use MegaSaM [2] to estimate camera parameters, depth maps, and optical flow between nearby frames. The key idea of our method is to efficiently construct all-to-all long-term 2D tracking by reusing optical flow estimated by MegaSaM. Using the resulting tracking data, Shape of Motion[4] renders N views from fixed locations to simulate multi-view video cameras. The generated videos are then fed to the 3D Video Loops algorithm [3] to produce the final looping 3D video. Figure 1 shows the proposed pipeline.

2.1 All-to-All 2D Tracking from Optical Flow

Shape of Motion requires long-term 2D tracking across all pairs of frames. However, the conventional pipeline relies on Boot-STAPIR [5], which performs tracking independently for each frame pair. This results in high computational cost and becomes a significant bottleneck when processing long video sequences.

To address this issue, we reuse the optical flow estimated by MegaSaM. MegaSaM does not estimate optical flows only between adjacent frames but also between frames separated by fixed temporal intervals $\Delta \in \{1, 2, 4, 8, 15\}$. We denote the optical flow mapping from frame at t to $t + \Delta$ by $\mathbf{F}_{t \rightarrow t+\Delta}$. Given a source frame at t and a target frame at s , we decompose the temporal offset into a sequence of available temporal steps $\{t_0 = t, t_1, \dots, t_K = s\}$, and corresponding optical flows are progressively composed as

$$\mathbf{x}_s = (\mathbf{F}_{t_{K-1} \rightarrow t_K} \circ \dots \circ \mathbf{F}_{t_0 \rightarrow t_1})(\mathbf{x}_t). \quad (1)$$

We find the shortest temporal path by tracing the largest possible steps, Δ , to avoid error accumulation of adjacent-frame tracking and to speed up processing. This reduces optical flow compositions and improves efficiency. The resultant all-to-all 2D tracking data are used as constraints for dynamic 3D scene reconstruction in Shape of

Table 1: Comparison of long-term 2D tracking methods for Shape-of-Motion [4] preprocessing. Time denotes the execution time of 2D tracking in seconds. All results are reported as mean $\pm$ standard deviation over four sequences.

Method	Time (s) $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BootsTAPIR [1]	3536 $\pm$ 2521	27.82 $\pm$ 2.69	0.877 $\pm$ 0.071	0.132 $\pm$ 0.071
Ours	28.8 $\pm$ 7.29	27.33 $\pm$ 2.40	0.873 $\pm$ 0.073	0.135 $\pm$ 0.071

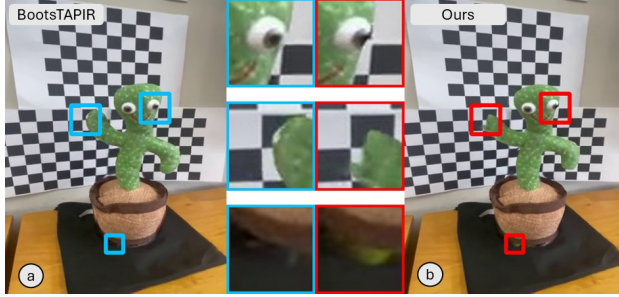


Figure 2: Qualitative comparison of dynamic 3D reconstruction results. (a) reconstruction using BootsTAPIR [1] for long-term 2D tracking. (b) reconstruction using the proposed tracking method. The center column highlights representative regions where qualitative artifacts or reconstruction failures are observed.

Motion, together with the camera poses and depth maps estimated by MegaSaM.

2.2 Dynamic 3D Reconstruction and Loop Rendering

Using the camera poses, depth maps, and all-pairs long-term 2D tracking, we reconstruct a dynamic 3D scene with Shape of Motion. The reconstructed scene is represented using 3D Gaussian Splatting, which enables efficient rendering.

To generate stable-view videos suitable for looping, virtual fixed cameras are placed in the reconstructed 4D (3D + time) space based on the input camera trajectory. For each virtual camera, rendering is performed along the temporal dimension, and the rendered videos are used as input to the 3D Video Loops algorithm [3] enabling the generation of temporally seamless looping 3D videos in the multi-plane video data format.

3 RESULTS AND DISCUSSION

We captured 4 monocular videos using an Apple iPhone 12 mini. One video was 3 seconds long, and the others were 4 seconds long. All videos were recorded at 1280 $\times$ 720 pixels. We captured repeating dynamic scenes, such as toys moving periodically and grass swaying in the wind. For each sequence, long-term 2D tracking was calculated using our approach, which reuses the optical flow estimated by MegaSaM [2]. Shape of Motion [4] was then executed using the resulting tracking data as input. We used a desktop PC equipped with an Intel Core i7-12700 (up to 4.9 GHz) CPU, 64 GB RAM, and an NVIDIA TITAN RTX GPU with 24 GB VRAM.

We compared our approach with BootsTAPIR [1], which is the standard 2D tracking method used in Shape of Motion. It is widely regarded as a state-of-the-art approach for tracking any point (TAP), enabling long-term dense tracking in monocular videos. We evaluated the final videos reconstructed by Shape of Motion, using the execution time of 2D tracking as well as peak signal-to-noise ratio (PSNR), structural similarity (SSIM), and learned perceptual image patch similarity (LPIPS) as evaluation metrics.

As shown in Table 1, the proposed method is able to construct long-term 2D tracking while largely preserving reconstruction quality by reusing the optical flow estimated by MegaSaM. Across all sequences, the PSNR, SSIM, and LPIPS values are comparable

to those obtained with BootsTAPIR. As illustrated in the close-up views in Figure 2, although slight degradations can be observed in regions with rapid motion or near the boundaries between dynamic objects and the background, no significant differences in overall visual quality are observed.

In terms of processing time, the proposed method achieves a substantial reduction compared to BootsTAPIR. This improvement comes from reusing optical flow estimated by MegaSaM instead of performing pairwise tracking.

Overall, these results demonstrate that the proposed method satisfies the requirement of all-pairs long-term 2D tracking in Shape of Motion, while significantly reducing preprocessing time without compromising reconstruction quality, making it a practical preprocessing solution for looping 3D video generation.

4 CONCLUSION

We proposed a pipeline for generating looping 3D videos from casually captured monocular videos recorded with a moving camera. By reconstructing a dynamic 3D scene and rendering it from virtually fixed viewpoints, we demonstrated the feasibility of generating looping 3D videos without requiring multiple physical cameras or controlled capture setups. Efficient long-term 2D tracking is a key enabling component that supports the generation of looping 3D video in our pipeline. By reusing the optical flow estimated by MegaSaM, our method efficiently constructs long-term 2D tracking for all frame pairs required by Shape of Motion. As a result, compared to the standard BootsTAPIR-based preprocessing approach, our method significantly reduces preprocessing time while maintaining a high quality reconstruction.

Visual artifacts and local reconstruction failures may still occur for fast-moving or complex objects. We also observe that the choice of virtual fixed camera viewpoints affects occlusions and the visibility of dynamic motions, which in turn influences the perceived loop quality. We further aim to improve the robustness of tracking and reconstruction in scenes with fast motion, and to systematically investigate virtual camera viewpoint selection strategies to achieve more stable and higher-quality looping 3D video generation.

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